

1. System Functionality

SYS-001

Description: The system shall have a 4 digit display to display a clock in the 24-hour format (e.g. 18:04).

SYS-002

Description: The system shall allow the user to be able to set the time in 2 different ways, manually or automatically.

SYS-003

Description: The system shall use the MCU:s internal RTC to keep track of real time.

2. Communication Functionality

SYS-011

Description: The system shall attempt to retrieve current time via the CAN-network on startup. If it is available, use it to set the MCU:s internal RTC.

SYS-012

Description: The system shall use SPI to communicate with and draw to the 4-digit display.

3. Hardware I/O and Display Functionality

SYS-021

Description: The system shall display the MCU:s current RTC on the 4-digit-display. The display shall be updated every second.

SYS-022

Description: The system shall allow the user to set the time manually by pressing the TIME-SLCT button. This shall pause the display updates and enter manual time set mode.

SYS-023

Description: When entering manual time set mode, the hour field shall be selected automatically. The user shall be able to increment the field by 1 each time the TIME-SET button is pressed. The user shall be able to switch to the minute field by pressing TIME-SLCT and be able to increment it by 1 every time TIME-SET is pressed.

SYS-024

Description: The system shall allow the user to exit the manual time set mode by pressing TIME-SLCT when the minute field is selected. This shall set the MCU:s internal RTC to the selected time, and resume updating the display every second.

SYS-025

Description: The system shall allow the user to manually retrieve time over the CAN-network by holding down the TIME-SLCT button for 2 seconds. If this is successful, the system shall update the MCU:s internal RTC.

SYS-026

Description: The system shall play a sound for 200ms using the buzzer every full hour (e.g. any time the minute field of the MCU:s internal RTC updates to 00).

Note: "pressed" means any individual rising-edge detection

Requirements mapping

Application: 021, 022, 023, 024, 025, 026

DisplayTaskApplication: 021, 022, 023, 024

Abstraction: 011, 021, 022, 023, 024, 025, 026

RTOS: 021, 022, 024, 025, 026

RTC Service: 011, 021, 024, 025, 026

COM Service: 011, 021, 024, 025, 026

Gfx Service: 021, 022, 023, 024

DHA: 021, 022, 023, 024

IoHwA: 022, 023, 024, 025, 026

